



NATTHAKRIT BENJAPATANAMONGKOL

B. Eng. in Smart Materials Technology and
Robotics and AI Engineering

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EDUCATION

2021 - 2024

King Mongkut's Institute of
Technology Ladkrabang

Smart Materials Technology and
Robotics and AI Engineering
(International Program)

Relevant Courses:

Deep Learning, Computer Vision,
Big Data, Cybersecurity
Capstone project - Parking
reservation system

GPA: 2.86

2017 - 2020

St.Paul's School Darjeeling, India

Science - Mathematics

TECHNICAL SKILLS

- AI & ML: Python, Langchain,
Tensorflow/Pytorch, OpenAI, YOLO
- DevOps: AWS Bedrock, Docker, Git
- Web: Next.js, HTML, CSS

SOFT SKILLS

- Project Management
- Teamwork
- Problem-solving
- Cross-Cultural Temawork- Japan Intern

LANGUAGES

- Thai (Native)
- English (IELTS overall band: 7)



PROFILE

Aspiring AI researcher with a strong academic foundation in robotics and AI applications, backed by a B.Eng. in Robotics and AI. Passionate about developing multi-agent intelligent systems and applying RAG-based architectures for human-AI collaboration. Seeking a Master 's degree position to conduct advanced research and develop next-generation AI systems focusing on Large Language Models (LLMs) and innovative generative AI



WORK EXPERIENCE

King Mongkut's Institute of Technology Ladkrabang

Freelance AI engineer

2025 - PRESENT

AI-Powered HR chatbot (Lead Developer)

- Architected and optimized a full-stack RAG pipeline using Python, AWS Bedrock, and a reranker model
- Developed and containerized the application with a Next.js, Python, and Docker.

Develop the chatbot for KMITL central lab (Project Lead)

- Lead 2 members team to design and built an optimized RAG pipeline with an integrated reranker to ensure high-accuracy responses from multi-source data (CSV files, website data).
- Developed system in Python using AWS Bedrock, delivering a Streamlit UI and image response capabilities

Kiddee Lab

Robotics Instructure

2025

- Designed and developed a new robotic curriculum on the VEX Robotics platform
- Instructed student on construction and programming using Tinkamo Robot Kits.

G-able Company Ltd.

Internship as AI engineer

July - October 2024

Develop the chatbot for ITSM department

- Developed a chatbot with a multi-agent-system using Langchain to optimize IT Service Management workflows
- Integrated SQL query tools, API data retrieval, and web search functionalities to enhance chatbot responses.

Fukuoka Institute of Technology, Japan

Internship program

June - July 2024

- Developed a racing simulation using Unity for research and training purposes.
- Designed and Programmed a line-tracking robot with embedded system and real-time processing.



PROJECTS EXPERIENCE

Emotion Classification

2025

- Create a system that receive text from speech to text model then classify text to emotion in 3 categories by using Openai GPT4 LLM model.

Parking Reservation System

2024

- Developed parking reservation system using a Rasberry Pi as the central controller
- Implemented real-time spot checks using ultrasonic sensors and automated vehicle verification via OCR on camera feeds.
- Built the user reservation interface with Next.js, which communicated with a backend to validate license plates.

Driver Drowsiness Detection System

2023

- Developed a real-time alert system using a Raspberry Pi and camera to monitor and alert drowsy drivers.
- Trained a custom YOLO model on a labeled dataset, achieving 97% accuracy in detecting visual signs of drowsiness.